ZBS | Združenje bank Slovenije
IzobraŽevalni center

Possibilities of LGD Modelling

Conference on Risk Management in Banks

François Ducuroir

Reacfin

Ljubljana, October 22, 2015

About Reacfin

Reacfin s.a. is a Belgian-based **actuary, risk & portfolio management consulting** firm.

We develop **innovative solutions and robust tools** for Risk and Portfolio management.

The company started its activities in 2004 as a **spin-off of the University of Louvain**, focused on actuarial consultancy to Belgian insurers, pension funds and mutual organizations. Rapidly, Reacfin expanded its business internationally and broadened its scope to various aspects of quantitative & qualitative risk management, financial modeling and strategic advice to financial institutions.

Spread over its 3 offices in Louvain-La-Neuve, Antwerp and Luxembourg, Reacfin employs about 30 consultants most of which hold PhD's or highly specialized university degrees.

Reacfin

What we do

- Modeling
- Risk implementation advisory
- Validation & model reviews
- Specialized strategic risk consulting

Reacfin s.a./n.v.
Place de l'Université 25
B-1348 Louvain-la-Neuve

Tel : + 32 (0)10 84 07 50
info@reacfin.com
www.reacfin.com

Reacfin
Know-How to Risk

Reacfin's 4 core fields of expertise

Risk & Portfolio Management

ALM, Portfolio Management & Quantitative Finance

- Implementation and calibration of stochastic models for valuation, trading and risk Management purposes
- Times series analysis & modelling
- Pricing of financial instruments & development of ALM models
- Design/review/implementation of systematic trading & hedging strategies
- Business intelligence in ALM or Portfolio Management
- Tools development (Valuation, Pricing, hedging, portfolio replication, etc.)
- Design of Capital Management solutions

Qualitative Risk Management, Restructuring & Operations

- Organization & Governance
- Businesses restructuring & change management
- Implementation and industrialization of processes
- Internal & regulatory reporting (KRI's & KPI's dashboards)
- Model Review frameworks
- Model Documentation

Insurance specialities

Life, Health and Pension

- DFA* Models
- Capital Requirement assessment
- Business valuation support
- Product development (pricing, profitability,..) & Reserving
- Model validation

Non-Life

- Reserving: triangle methods, individual claims modelling
- Pricing: frequency and severity modelling, large claims analysis, credibility methods, commercial constraints
- DFA models: cash-flows projection, calibration of models
- Reinsurance: modelling covers, optimal reinsurance programs

Possibilities of LGD Modelling

Agenda

- Some key theoretical concepts and their implications

- Practical aspects of model implementation

Reacfin
Know-How to Risk

Determining the cost of Credit Risk

Expected Loss (EL)	=	Probability of Default (PD)	X	**Loss Given Default (LGD)**	X	Exposure at Default (EAD)
Definition		The likelihood of a borrower being unable to repay		The fraction of exposure at default that is lost in the case of default		The exposure at risk in the case of default
		The chance that an event of default will occur over a given period of time		*the complement of the Recovery Rate (RR): LGD=1-RR i.e. Degree of security of a facility, expected percentage of the EAD that will be lost, after default, representing the total economical loss*		*The maximum potential loss on principal, i.e. initially provided principal amount, minus the portion that effectively has been used already to cover claims, before the default point.*

ZBS | Združenje bank Slovenije
IzobraŽevalni center

Long-Run LGD & Down Turn LGD

"Expected Loss" (EL)

- Anticipated average **annual loss rate**
- **Foreseeable 'cost' of doing business**
- **Not 'risk'** as investors think of it, but rather a charge which affects anticipated yield

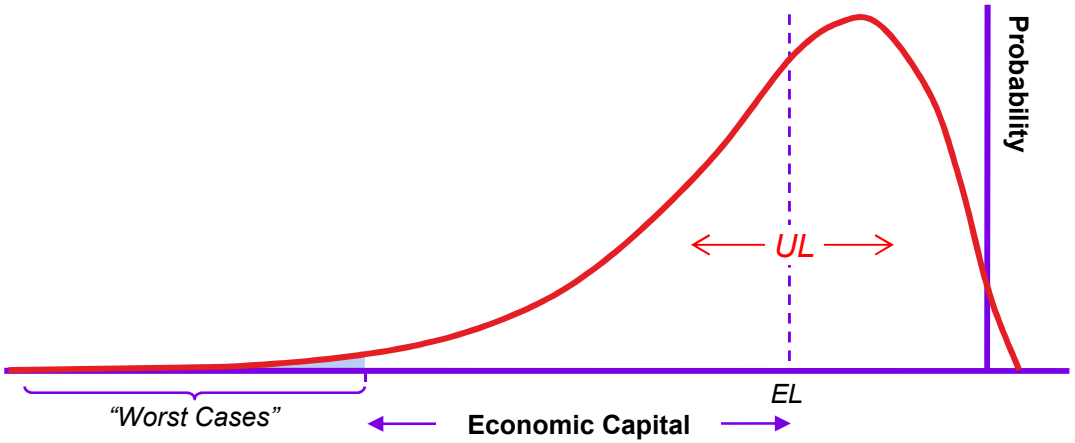
"Unexpected Loss" (UL)

- Anticipated volatility of loss rate (or value) i.e. **volatility of EL**
- Results in **volatility of return** over time
- Unforeseeable but **inevitable** (distressed conditions)
- Requires a balance sheet cushion of 'economic' capital

Price into Products
Based on Long-Run LGD (average through the cycle)

Hold Capital
Based on Down-Turn LGD (LGD in adverse market conditions)

Losses distribution



Reacfin
Know-How to Risk

Typical benefits of improved LGD models

- **Better assessment the real risks** of the loan books: support fundamental management decisions with regard to the business development, commercial policy and the corporate strategy

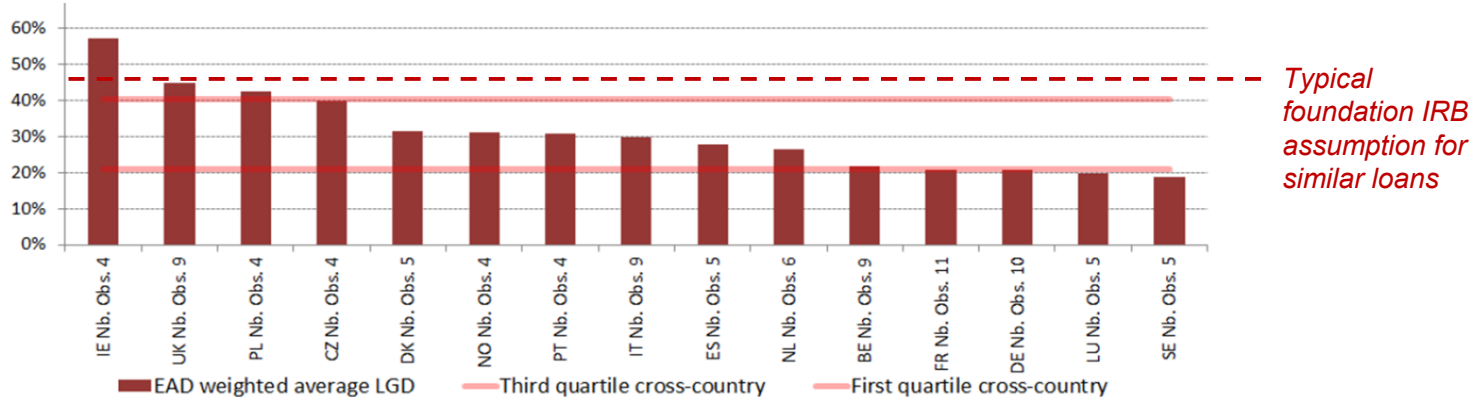
- **Positive P&L Impact:** Accounting for realistic expected losses in the loans price-setting and enabling better discrimination between loans types, debtors, etc..

  For instance, per € 1Bn EAD, a 10% reduction in LGD induces (at same pricing level) a P&L impact:
  - *For BB- rated debtors (i.e. 1Y-PD~2-3%) of about €2-3Mn/Y*
  - *For B rated debtors (i.e. 1Y-PD~3-5%) of about €3-5Mn/Y*

- **Better assessment of capital requirements** (i.e. improved estimation downturn LGD, taking into account the specificities of the book).

  Peers countries comparison (see graph below for SME loans) shows that in many European countries (Except Ireland), Internal model LGD's proved in-Line (UK & PL) or materially lower than that assumed by standard/foundation approaches

EAD-weighted average LGD, non-defaulted IRB exposures, SME, by country



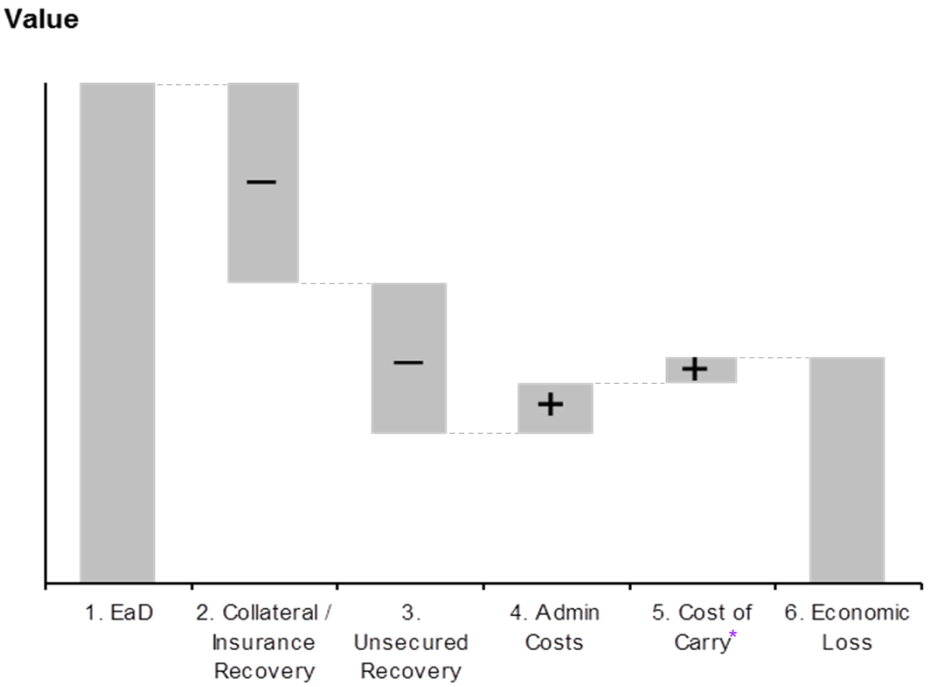
Source: European Banking Authority, Dec 2013

Reacfin
Know-How to Risk

LGD is a facility & asset specific measure

$$\text{LGD} = \frac{\text{Economic Loss}}{\text{Exposure at Default (EAD)}}$$

Example Economic Loss Calculation Steps for Loans

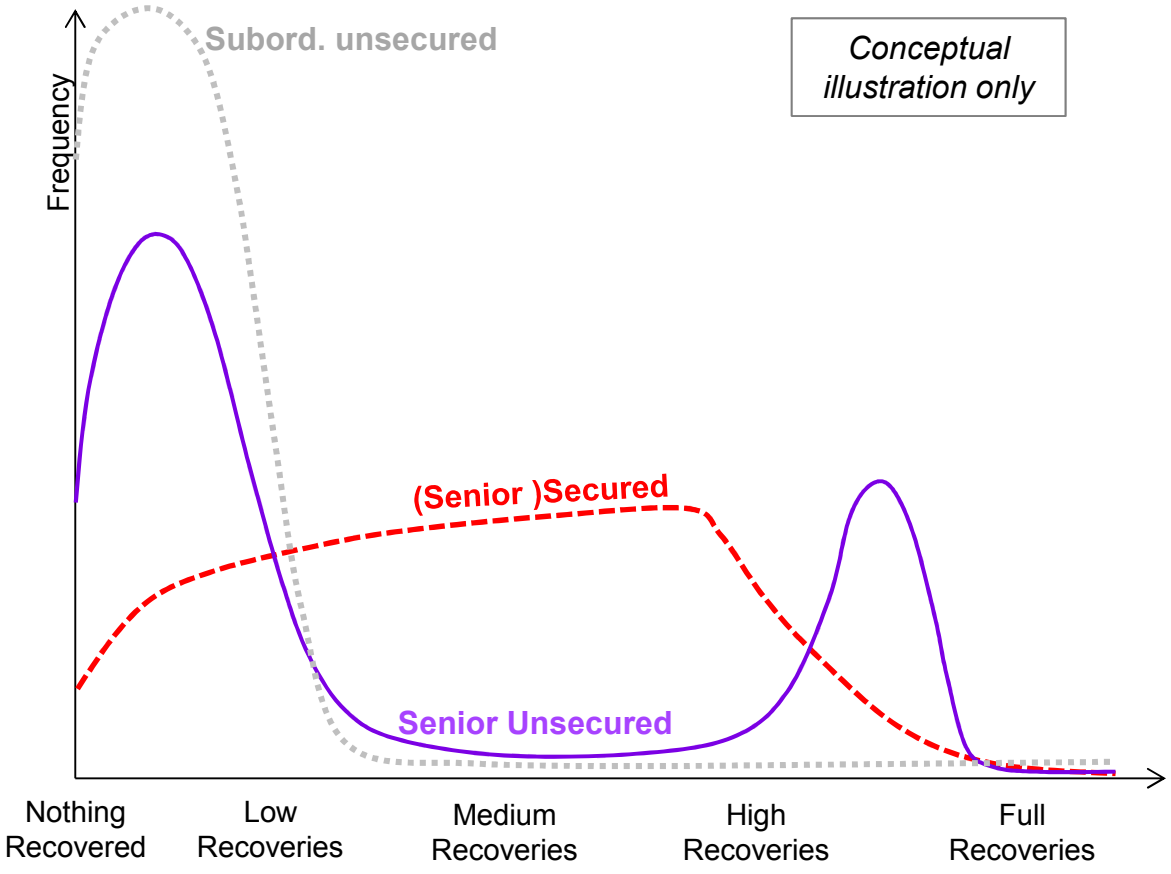


- Conceptually, LGD represents the exposure, net of recoveries, **lost in a default**
- LGD strongly depends on the **seniority** of exposure, type of **collateral** and **borrower**
- For compliance with the **Advanced IRB** approach of Basel 2, the calculation of LGD must be based on a model calibrated through a **statistical analysis of historical default** experience
  - Internal records on both cash and asset recovery are the foundation
  - Both direct (legal, repossession) and indirect (collections department) costs are included
  - Granular differentiation between different types of borrowers, structures, collateral,…
- Historic external data is often not readily available. Frequently internal data will need to be beefed-up by substantiated, carefully structured/systematized and documented **expert judgment.**

(*) Interest rates charges related to the fact that the recovery process takes time

ZBS | Združenje bank Slovenije
IzobraŽevalni center

Typically observed distributions of recoveries



Senior Unsecured debt recoveries distribution often prove to be **bimodal** ➔ Challenge in adequately modeling the LGD using known (usual mono-modal) distributions (e.g. Beta distribution)

Senior Secured debt recoveries on the other side may prove much more **evenly distributed** in certain intervals given the type of collateral

Unsecured subordinated while most extreme may prove having best shape **match with traditional theoretical assumptions**

Basel II/III requirements to own-LGD estimates

Source: **BCBS 128 - Basel II: International Convergence of Capital Measurement and Capital Standards: Revised Framework - Comprehensive Version**

- Art.468 **"Estimate an LGD for each facility that aims to reflect economic <u>downturn</u> conditions ".**
  - observed during periods of high credit losses, **forecasts** based on appropriately conservative assumptions, or other similar methods
  - using either **internal and/or external** data.
  - Supervisors will (…) **encourage** the development of such appropriate approaches (…).

- Art.469 "Consider the extent of any **dependence between the risk of the borrower and that of the collateral** or collateral provider."

- Art.470 "LGD estimates must be grounded in **historical recovery rates** and, when applicable, must not solely be based on the collateral's estimated market value."

- Art.471 "LGD (…) should reflect (…) that the bank would have to recognize additional, unexpected losses **during the recovery period**."

- Art.472 "LGD must be based on a **minimum data observation period** that should ideally cover at least **one complete economic cycle** but must in any case be **no shorter than (…) seven years** for at least one source."

What about the Capital Requirements floor?

Consultative Document

Basel Committee
on Banking Supervision

Consultative Document

Standards

Capital floors: the design of a framework based on standardised approaches

Issued for comment by 27 March 2015

December 2014

BANK FOR INTERNATIONAL SETTLEMENTS

- **Objectives:**
  - "reduce variations in capital ratios across banks"
  - "reliability and comparability of risk-weighted capital ratios"
  - "strengthen the link between the standardised and IRB approaches"
- **Why?**
  - "Extremely low levels of internally modelled RWAs have been observed for some exposure categories."
- **Possible Approaches**
  - "applied to each major risk" (risk category-based floors)
    - Through adjustments on numerator of the capital ratio
    - By adjusting RWA's
  - "based on total RWAs" (aggregate RWA-based floor)

Could it make A-IRB approaches less attractive for banks?
Does it on the contrary trigger more demand for advanced models?

Main types of LGD assessment models

- **Work-out LGD's**
  - Based on default & recovery work-out status information (I.e. assessment is differentiated for each step of the recovery process) and can be enriched using financial statement information

- **Market LGD's**
  - Based on the residual value of defaulted market debt instruments as observed after their default

- **Implied Market LGD's**
  - Based on Credit Spreads trading on financial markets

- **Implied historical LGD's**
  - Long term historical cohorts observed averages for comparable instruments (typically only used for retail loans or for back-testing purposes)

Work-out LGD models

- Derived from the set of estimated **projected cash flows along the different recovery states** ("clustering of the default & recovery trajectory "):
  - Based on historical recovery process information
  - Discounted at a rate adequately reflecting the borrowers risks as of the date of default
  - Accounting for collateral pledging and expenses (direct/indirect)
  - Estimated for each asset type in the bank & **segmented along debtors or loans characteristics** *(e.g. rating, maturities, etc.)*

- Standard market practice distinguish **3 work-out LGD** *(for assets sensitive to the cycle)*
  - Current LGD (**CLGD**): Current best estimate given prevailing market conditions
  - Long-Run LGD (**LRLGD**): average long-term LGD corresponding to an a-cyclical scenario *(used to calculate "expected losses")*
  - Downturn* LGD (**DLGD**): LGD at the worst time of the economic cycle *(used to calculate "unexpected losses).*

- **Basic clustering of trajectory principles**:

  LGD is a probability weighted sum of clustered outcomes in present Value: $LGD_{total} = \sum_{i=1}^{n} P(State_i) * PV(LGD_i)$
  - Cured* vs. uncured states ➔ bimodal segmentation
  - Intensive Care & Recovery risk mitigation techniques impact (e.g. restructuring** & back to cured state)
  - Collateral liquidation accounted for on 'unsecured LGD to compute 'secured LGD'
  - All calculations must be done in present value given that recovery process may sometimes be very time consuming

Sources: European Banking Association, BCBS, Reacfin
(*) i.e. after some time, the debtor recovers and starts reimbursing as expected
(**) e.g. allowing longer reimbursement period or reducing the due amount

Pro's & Con's of Work-Out LGD models

- **Advantages**
  - Most **common** model in banks – Applicable to any debt commitment
  - **Transparent** to regulator, and aligned with Basel definition of LGD
  - Conceptually relatively **easy to understand**
  - Most **bank-specific** method of all, as long as there is sufficient historical data available to make reliable estimates

- **Disadvantages**
  - Difficult to build-in **some forward looking character.** (Requires often unavailable data covering the full credit cycle and sufficiently reliable **unbiased** expert judgement)
  - A suitable **discount rate** must be selected, consistent with the ex ante risk & forward looking
  - Bankruptcy claims are often not settled only in cash but with **collateral of different nature** but with no secondary market When performing statistical assessments is then **critical to be able to relate recovered amounts to realized collateral**
  - In portfolios with a **low rate of defaults** (low default portfolio, or LDP), there are often **insufficient historical experience** to make a reliable estimate using the Workout LGD method, so external proxies may have to be considered (for which relevance may, in cases, prove questionable)

Critical: **Data Quality granular & exhaustive enough to account for the different steps in the Work-Out process**

Implied Market LGD

- **Concept**
  - Recovery rate derived from traded credit spreads
    - Investors already account for the expected loss in the market spreads they require for traded debt instruments.
    - Therefore, if one knows the PD's (e.g. usual by using structural models of more simply by approximating them using ratings),
    - LGDs can be derived from risky (but not defaulted) bond prices or other traded credit default instruments (e.g. Credit Derivatives).

- **Advantages**
  - Inherently forward looking
  - Gives quite good responsiveness of LGD evolution, despite all idealizations in assumptions (ok as early warning)
  - Typically used if no or too little historical data is available (low-default portfolios)

- **Disadvantages**
  - Advanced Implied Market models often require material modeling assumptions
  - Requires adjustments & modeling assumptions for obtaining a through the cycle (TTC) LGD outcome

- **<u>An approximate simple model (Hull Formula)</u>**

$$(1+i) = (1+r) * (1- PD) + (1+r) * PD * RR$$

where r= risky rate, i=risk-free rate , PD is the cumulated Probability of Default and RR= Recovery Rate

$$\Rightarrow \quad RR = [(1+i) - (1+r) * (1- PD)] \;/\; [(1+r) * PD]$$

Implied Market LGD

Example for the Hull approximation formula

FORMULA: $(1+i) = (1+r) * (1 - PD) + (1+r) * PD * RR$

where r= risky rate, i=risk-free rate , PD is the cumulated Probability of Default and RR = Recovery Rate

5y Swap Rate	2,00%	2,00%	2,00%	2,00%
Corporate Bond Rate	6,00%	7,00%	13,60%	2,00%
Bond Maturity	5y	5y	5y	5y
Bond Rating	Ba	Ba	Ba	Ba

Cum. PD	10,215%	10,215%	10,215%	10,215%

RR	63,1%	54,3%	0,0%	100,0%
LGD	36,9%	45,7%	100,0%	0,0%

Any rate above 13,60% would demonstrate market expects higher default rate than PD assumption

No credit spread ➔ assumes no expected loss due to defaults

Average cumulative default rates (%), 1970–2006.

Term (years):	*1*	*2*	*3*	*4*	*5*	*7*	*10*	*15*	*20*
Aaa	0.000	0.000	0.000	0.026	0.099	0.251	0.521	0.992	1.191
Aa	0.008	0.019	0.042	0.106	0.177	0.343	0.522	1.111	1.929
A	0.021	0.095	0.220	0.344	0.472	0.759	1.287	2.364	4.238
Baa	0.181	0.506	0.930	1.434	1.938	2.959	4.637	8.244	11.362
Ba	1.205	3.219	5.568	7.958	10.215	14.005	19.118	28.380	35.093
B	5.236	11.296	17.043	22.054	26.794	34.771	43.343	52.175	54.421
Caa–C	19.476	30.494	39.717	46.904	52.622	59.938	69.178	70.870	70.870

Issue with the method: How reliable are the PD's assumptions?

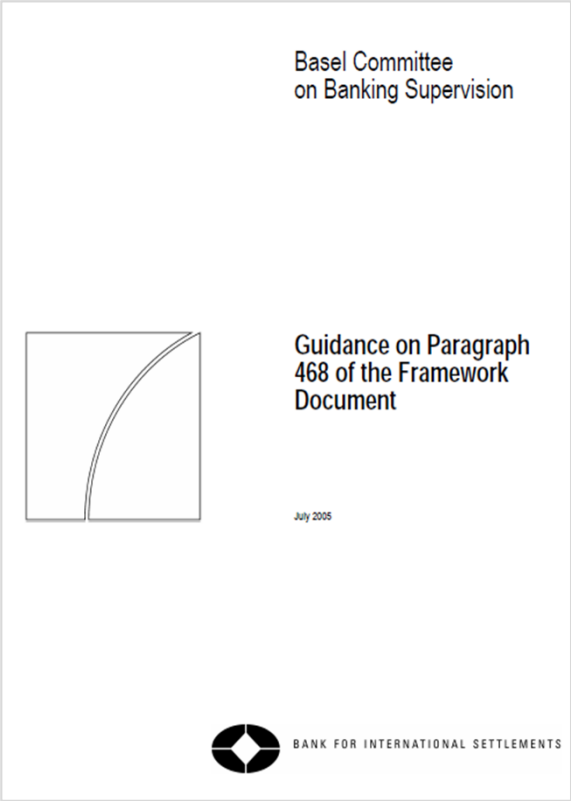
Typical model choices

- The selection of model drivers (e.g. Loans specificities [*type, size, covenants, etc.*], debtors profile [*size, sector, rating, geographies, etc.*], collateral & support, etc.) can affect capital requirements in excess of 50%.
- Hence, European banks' best practices aim at considering **sufficiently granular segments/clustering** to capture relevant loan distribution while keeping sufficient level of **homogeneous aggregation** to enable assessment of tail risks.
- The choice between the approaches is in practice mainly data driven, namely input data availability, which results in following scheme.

Sources	Input Data	Type of facilities		Mostly applicable to
		Defaulted facilities	Non defaulted facilities	
Market Values	Price Differences	(Explicit) Market LGD		Large corporate loans, listed bonds, sovereigns and banks
	Market Quotes (Credit spreads, equity prices, bonds, CDS, …)	*Often hardly applicable*	Implied market LGD	
Recovery and Cost Experience	Discounted Cash Flows	(Explicit) Workout LGD		Retail, SMEs and Large Corporates
	Historical Total Losses and estimated PD	Implied Historical LGD		Retail

Down-turn LGD

Guidance Paper


Basel Committee
on Banking Supervision

Guidance on Paragraph
468 of the Framework
Document

July 2005

BANK FOR INTERNATIONAL SETTLEMENTS

1. The potential for realized **recovery rates to be lower than average during times of high default rates** may be a material source of unexpected credit losses for some exposures or portfolios. Failing to account for this possibility risks understating the capital required to cover unexpected losses.

2. **Data limitations** pose an important challenge to the estimation of LGD parameters in general, and of LGD parameters consistent with economic downturn conditions in particular.

3. There is currently **little consensus** within the banking industry with respect to appropriate methods for incorporating downturn conditions in LGD estimates. A significant body of academic and practitioner research on this issue has Shown a wide disparate range of results concerning the potential impact of downturn conditions on LGDs.

The Committee has determined that a principles-based approach to elaborating on the requirements of paragraph 468 is most appropriate at this time.

DLGD principles: Down-turn conditions

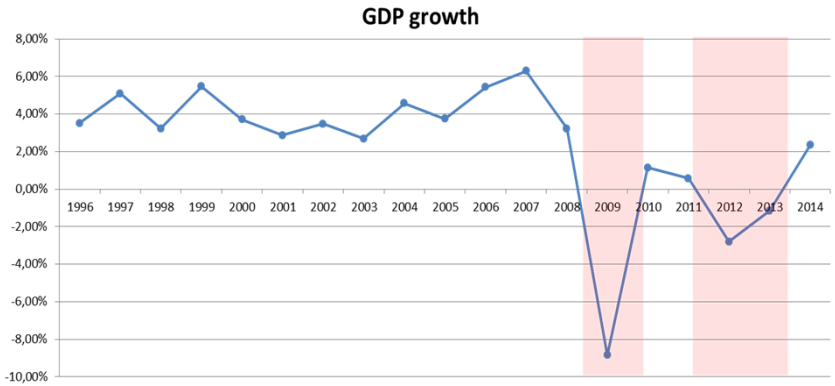
Principle 1

Identification of appropriate downturn conditions for each supervisory asset class within each jurisdiction.

- **Appropriate downturn conditions might be characterized, for example, by the following:**
  - For a well diversified wholesale portfolio, periods of **negative GDP growth** and **elevated unemployment** rates.
  - Periods in which observed historical **default rates have been elevated** for a portfolio of exposures that is representative of the bank's current portfolio.
  - For exposure where **common risk drivers** (e.g. collateral values) influence the default rates and the recovery rates, periods where those drivers are expected to be distressed.

Sources: *BCBS – Guidance on the paragraph 468 of the Framework Document, July 2005*

Slovenian Economy



DLGD principles: Adverse dependencies

Principle 2

Identification of adverse dependencies, if any, between default rates and recovery rates.

- Those adverse dependencies might be identified, for example, by some or all of the following:
  - A comparison of **average recovery rates with recovery rates observed during appropriate downturn periods** identified according to principle 1.
  - A statistical analysis of the relationship between observed **default rates and observed recovery rates** over a **complete economic cycle**.
  - For secured exposures where default is shown to be highly correlated with collateral values:
    - A comparison of recovery rate forecasts derived from robust statistical models that use both "typical **assumptions about collateral value** changes and appropriate "downturn" conditions identified according to principle 1.
    - A comparison of observed **recovery rates for defaulted exposures given typical collateral values** with those observed under conditions identified according to principle 1 where collateral values are depressed.
  - Identification of the **underlying factors (risk drivers) that determine recovery rates** and analysis of the relationship between those factors and default rates, combined with an assessment of the net impact of those factors on recovery rates under "downturn" conditions.

Sources: *BCBS – Guidance on the paragraph 468 of the Framework Document, July 2005*

DLGD principles: Dependencies incorporation

Principle 3

Incorporation of adverse dependencies, if identified, between default rates and recovery rates so as to produce LGD parameters for the bank's exposures consistent with identified downturn conditions.

- For example, for those exposures for which adverse dependencies between default rates and recovery rates have been identified through analysis consistent with principle 2, the **LGD estimates may be based on averages of observed loss rates during downturn periods** identified according to principle 1 or they may be derived from forecasts based on stressing appropriate risk drivers in a manner consistent with downturn conditions identified according to principle 1.

- If **no material adverse dependencies** between default rates and recovery rates have been identified through analysis consistent with principle 2, **the LGD estimates may be based on long-run default-weighted averages** of observed loss rates or they may be derived from forecasts that do not involve stressing appropriate risk drivers.

Sources: *BCBS – Guidance on the paragraph 468 of the Framework Document, July 2005*

Possibilities of LGD Modelling

Agenda

- Some key theoretical concepts and their implications
- Practical aspects of model implementation

Reacfin
Know-How to Risk

Architecture of typical LGD models

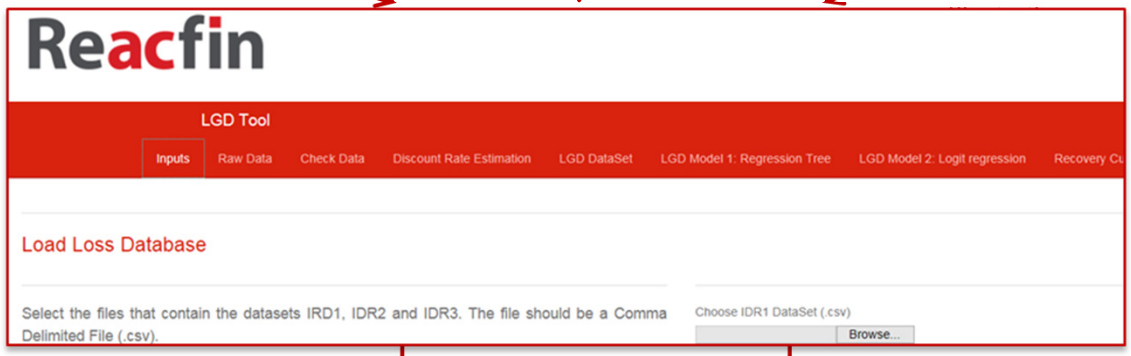
INPUT

Historical Information on defaulted loans*

*Historical loans pricing information***

*Market, Financial & macro-economic information***

LGD (Average & Down-turn) Assessment & Calibration tool



OUTPUT

LR-LGD assessments

DLGD estimations

(*) typically:
- Exhaustive loans specifications,
- Recovery cash-flows [incl. expenses],
- Related collateral & collateral proceeds)

(**) typically used as input to asset discounting rates for recovery cash-flows

(***) typically:
- Market information if available (Bonds quotes, ASW & CDS spreads, Equity prices)
- Macro-economic indicators (e.g. GDP growth, Inflation unemployment)
- Financials from annual accounts (e.g. leverage, total assets, etc.)

Calculating observed work-out LGD's

- The LGD for workout period i and **loan ID *j*** is calculated as followed:

$$LGD(i,j) = 1 - \frac{R_{discount}^{cumul}(i,j) - E(j)}{EAD(j)}$$

Where

- $R_{discount}^{cumul}(i,j)$ *are the* ***Cumulative Discounted Recoveries*** *for ID* j *and workout period* i*;*
- $EAD(j)$ *is the* ***Exposure-at-Default*** *for loan ID* j*;*
- $E(j)$ *are the* ***Expenses*** *related to the recovery process of loan ID* j*.*

- For loans the **EAD** for loan ID j is estimated as the on-balance sheet EAD. For guarantees and credit cards the final EAD is estimated as the on-balance sheet EAD plus the off-balance sheet exposure times a Credit Conversion Factor (CCF) which it is critical to estimate correctly based on actual observations (theoretical assumption may lead to material model flaws)

- **Discounting rate** should account for debtors actual risk profile. Typical starting point proxy are to consider
  - the original rates (or spreads) prevailing at loan's inception
  - the commercial target rate for a fixed rate loan with lowest rating (or defaulted) if priced and assuming loan term relevant for the expected average recovery process (e.g. 12 months).
  - Accounting for relevant penalty rates on guarantees & credit cards

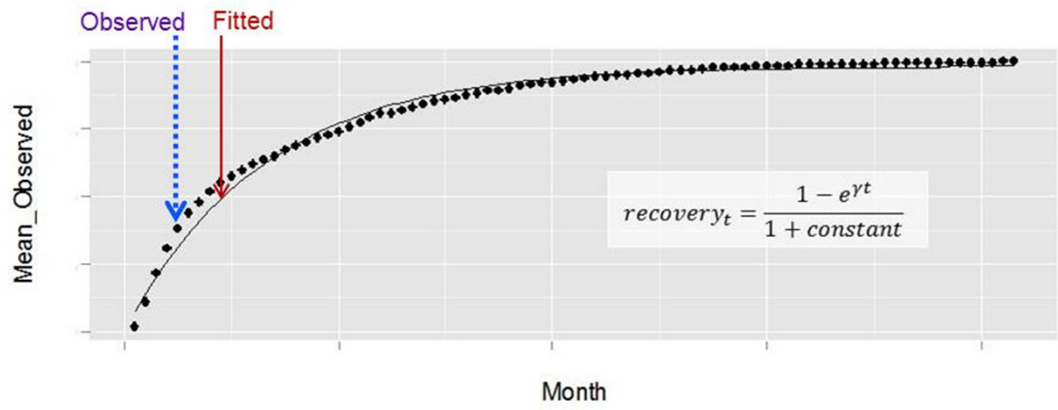
- **Expenses** may need to be estimated using regression techniques (often simple linear regressions) with threshold levels for smaller loans. Objective here is to correctly account for both variable costs directly related to identifiable cash-flow recoveries and fixed costs related to the overall work-out process.

- **High gain question: At which level** should we consider the LGD model (debtor level or loan level?). Answer depends on available data quality/granularity.

Enriching data with unclosed files information

- Sensu stricto, the historical LGD's should be calculated on the basis of closed files only (for which all recovery cash-flows are known).
- As already stated, the challenge of limited data availability may however materially impact the quality/reliability of LGD estimates.
- To partly overcome this issue, some European banks are enriching their data by taking into account information from unclosed files through regression techniques (typically exponential parametric fitting techniques)
- Adequate criteria's for the accounting or not of fitted proxies must be carefully set & documented

Illustrative example



ZBS | Združenje bank Slovenije
IzobraŽevalni center

Categorization of work-out LGD's

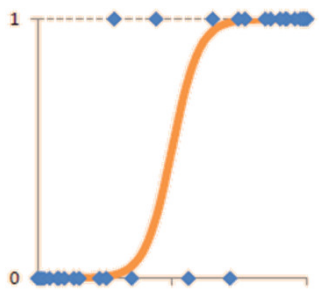
- Statistical inference technique are then used to categorize predicted LGD's given the loans characteristics.
- A common approach consists in the **segmentation** of the calculated expected LRLGD using analytical approaches:
  - **logistic regressions**
    - i.e. a regression model where the dependent variable is categorical (Yes or No)
    - Thus usually regressed on exponential functions
    - May prove pretty sensitive to input data and thus providing less stable results
  - **regression trees models**.
    - Regression tree algorithms produce a decision tree for the dependent variable by recursively partitioning the input space based on a splitting criterion, e.g. a weighted reduction of the within-node variance.
    - This approach is part of the common toolkit for LGD Modeling. See for example
      - Loterman G. et al. (2012), Benchmarking regression algorithms for loss given default modeling;
      - Bastos J. (2009), Forecasting bank loans loss-given-default;
    - A wide variety of decision tree algorithms exists in the literature. Most commonly used quantitative approach include
      - Conditional inference tree *(issue: sometimes less stable – more sensitive to input data)*
      - Recursive partitioning algorithm.
  - More advanced methods may include **Random Forest** algorithms or even **Neural Networks*** approaches

(*) While we have seen larger banks considering the approach we have never seen it put in production

Description on of analytical approaches

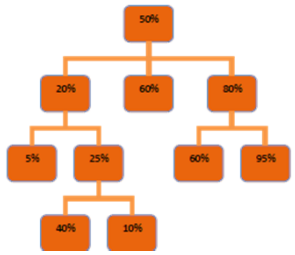
Logistic regression

- A Regression Technique designed for modeling the chance of the occurrence of an event.
- This produces individual scores for each debtor/loan based on the predictor variables.



Decision Trees

- Categorizes the occurrence chance of an event
- Producing individual scores for debtors / loans categories based on the predictor variables.
- Being categorical makes results easier to use in banking processes



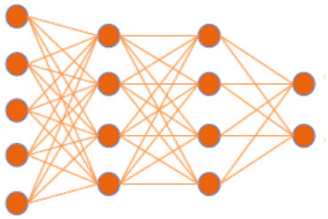
Random Forest

- Within a Random Forest model, a specified number of Decision Trees are build (>100) and regressing across them
- The conclusions of this technique are more difficult to implement for banks (no description of the target group rather different results for each loan/debtor

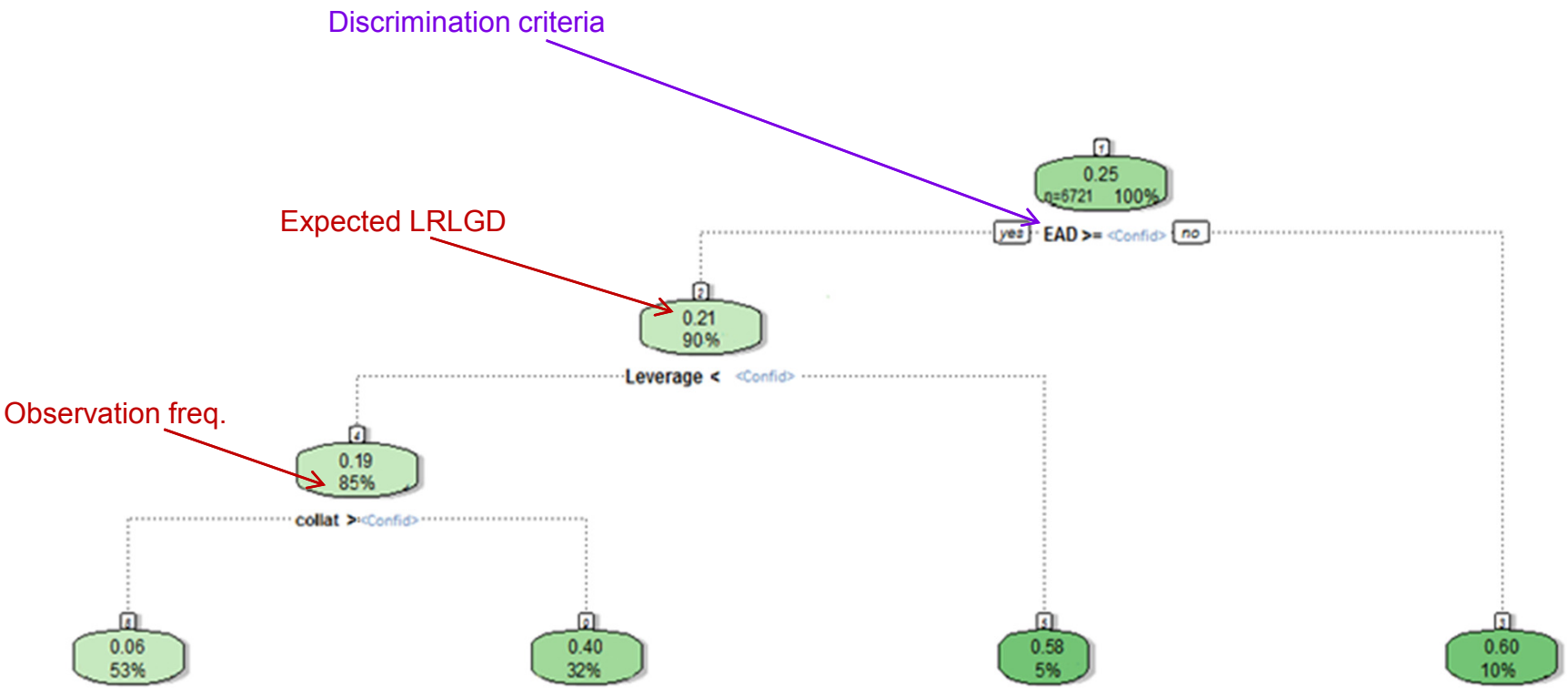
Neural Networks

Emerging technique being tested – No known "real-life' use in prodction

- Machine Learning data mining technique based layers of neurons to suggest a decision.
- The model is "trained" using using predictor variables and test cases (known results).
- Hard to interpret and sometimes often seen as a black box.



Example of tree using Recursive partitioning algorithm



Categorization approach facilitates translation to loans price setting mechanisms

ZBS | Združenje bank Slovenije
IzobraŽevalni center

Q&A

Thank you for your attention.

Any questions now?

For later questions:

Mail: Francois.ducuroir@reacfin.com
Tel: +32 472 32 05
Web: www.reacfin.com

ZBS | Združenje bank Slovenije
IzobraŽevalni center